[College / University Name] · [Department Name]

# SmartDock

Dock Appointment Scheduling & Yard Management
for Warehouses and Distribution Centres

**Student** [Student Name] ([Register Number])
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**Programme** [Programme, e.g. MBA / B.Tech], [Department Name]
**Academic year** 2026–2027

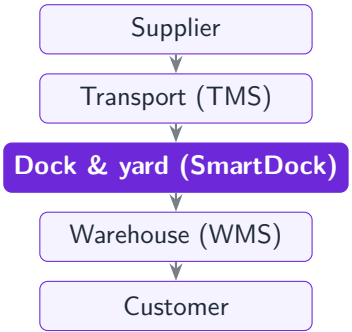
Supply Chain Management – Final Project

# Outline


1. Introduction: IT in supply chains
2. Problem statement
3. Objectives
4. Scope and limitations
5. Background and literature survey
6. Gap and proposed solution
7. System architecture
8. DFD, use case and ER diagrams
9. Technology stack
10. Modules
11. Implementation details
12. Screenshots and live demo
13. Testing and validation
14. Results and discussion
15. Benefits and applications
16. Conclusion
17. Future enhancements
18. References

# Introduction: Role of IT in Modern Supply Chains

- Supply chains compete on coordination between independent partners — suppliers, carriers, warehouses.
- IT provides:
  - **Visibility** — one shared, real-time picture
  - **Synchronisation** — partners work to one plan
  - **Automation** — rules replace phone calls
  - **Measurement** — KPIs for continuous improvement
- ERP, WMS, TMS, YMS . . . but the loading dock — the hand-over between transport and warehouse — is often still managed by phone and whiteboard.



# Problem Statement

Without dock appointments:

- Trucks arrive at the same peak hours → yard queues, idle docks at other times
- Driver detention and unplanned dock labour
- Double-booking and wrong doors (reefer, hazmat)
- No record of who is late or never arrives
- No data: utilisation? busiest hours? turnaround?

## Problem

Build a web platform where warehouses publish dock capacity and carriers self-book **compatible, conflict-free** appointments, while yard events are captured to measure and improve **utilisation, punctuality and turnaround** — securely, for many warehouses and carriers.

# Objectives

1. Multi-tenant platform with per-warehouse carrier approval
2. Configurable scheduling: hours, 15–90 min slots, buffers, holidays, maintenance
3. Match loads to docks: inbound/outbound, trailer type, hazmat
4. Guarantee slot capacity under concurrent requests
5. Recurring bookings, rescheduling, automatic waitlist
6. Digitise yard execution; detect late arrivals and no-shows
7. Carrier reliability score and operational KPIs
8. Audit trail and notifications (in-app, e-mail, SMS)
9. Cloud deployment, automated tests, simulation study

# Scope and Limitations

## Scope

- Users: warehouse managers, carriers, super admin
- Full appointment lifecycle, waitlist, yard events, KPIs, audit, notifications, exports
- One cloud service + managed database, any browser
- Demo data: 2 DCs, 5 typed docks, 6 carriers, 5 weeks

## Limitations

- No ERP/WMS/TMS integration (PO/ASN entered manually)
- Gate events recorded manually (no ANPR / IoT)
- Fixed slot length per site
- Super-admin role has no UI yet
- Evaluation on generated / simulated data
- Free hosting: cold start, small CPU

# Background: Docks, Yards and Dock KPIs

- **Dock door**: inbound (receiving) or outbound (shipping); capacity per period
- **Equipment**: dry van, reefer (cold chain), flatbed, container; hazmat needs certified bays
- **Yard**: where trucks wait before / after docking
- **Appointment**: a reserved door for a time window

## Key performance indicators

- Dock utilisation = booked ÷ offered truck-slots
- On-time arrival rate
- No-show rate
- Turnaround = yard wait + dock time

# Literature Survey and Existing Systems

- Port **truck appointment systems** reduce congestion when enforced (Giuliano & O'Brien 2007; Huynh & Walton 2008; Zehendner & Feillet 2014; Phan & Kim 2015)
- Receiving/shipping and **cross-dock scheduling** drive warehouse throughput (Gu et al. 2007; Boysen & Fliedner 2010)
- **Write skew** can double-book under concurrency (Kleppmann 2017)

| | Phone / spreadsheet | Port TAS | Dock-scheduling SaaS | WMS/YMS suite | **SmartDock** |
|---|---|---|---|---|---|
| Carrier self-service | No | Yes | Yes | Partial | **Yes** |
| Capacity enforced | Manual | Yes | Yes | Yes | **Atomic, multi-server** |
| Dock-type / hazmat | Manual | Limited | Varies | Yes | **Yes** |
| Waitlist / recurring | No | Varies | Varies | Varies | **Yes** |
| Scorecard & KPIs | No | Terminal KPIs | Yes | Yes | **Yes** |
| Cost | Effort | Port-run | Subscription | High | **Free, open source** |

# Gap in Existing Systems and Proposed Solution

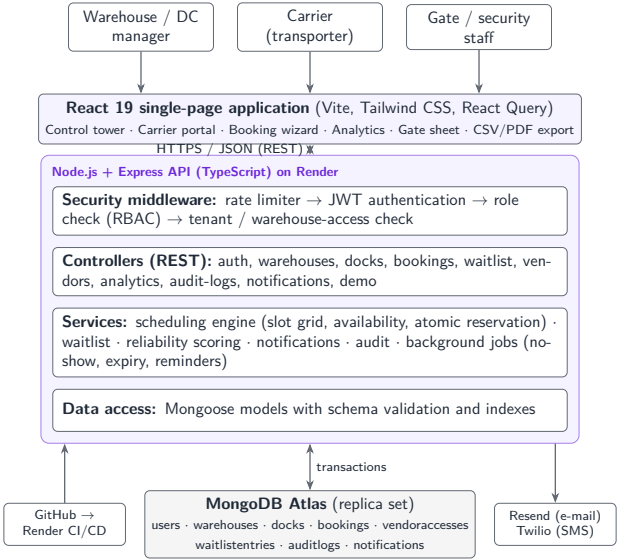
## Gap

- SMEs stuck between manual methods and expensive enterprise suites
- Port systems are not built for multi-tenant warehouses with mixed docks
- Simple calendars ignore compatibility, capacity and concurrency
- Booking separated from execution → no KPIs, no accountability

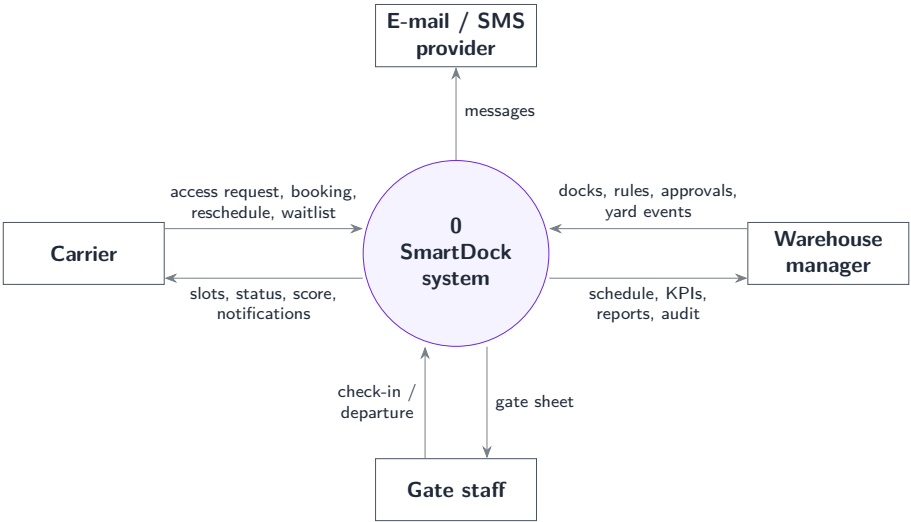
## SmartDock

- Per-warehouse rules and carrier access
- Only compatible docks and valid slots; concurrency-safe capacity
- Full lifecycle: request → approve → gate → dock → depart; reschedule, series, waitlist
- KPIs, reliability score, reports, audit, notifications
- Free, open source, one-file cloud deployment

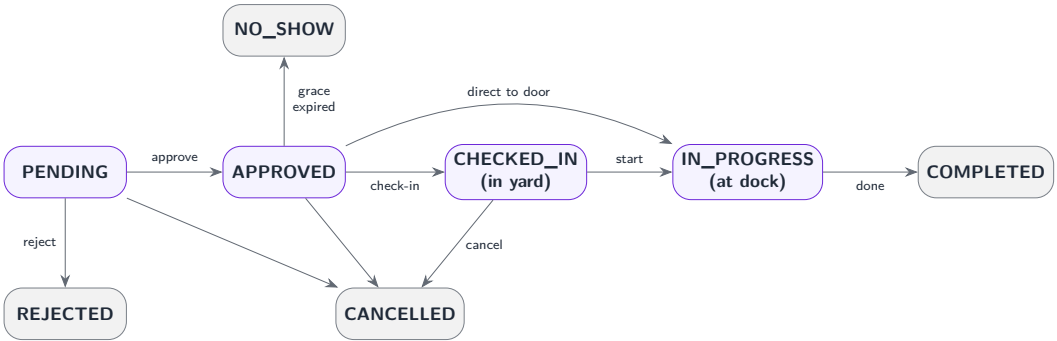
# System Architecture



# Context Diagram (DFD Level 0)

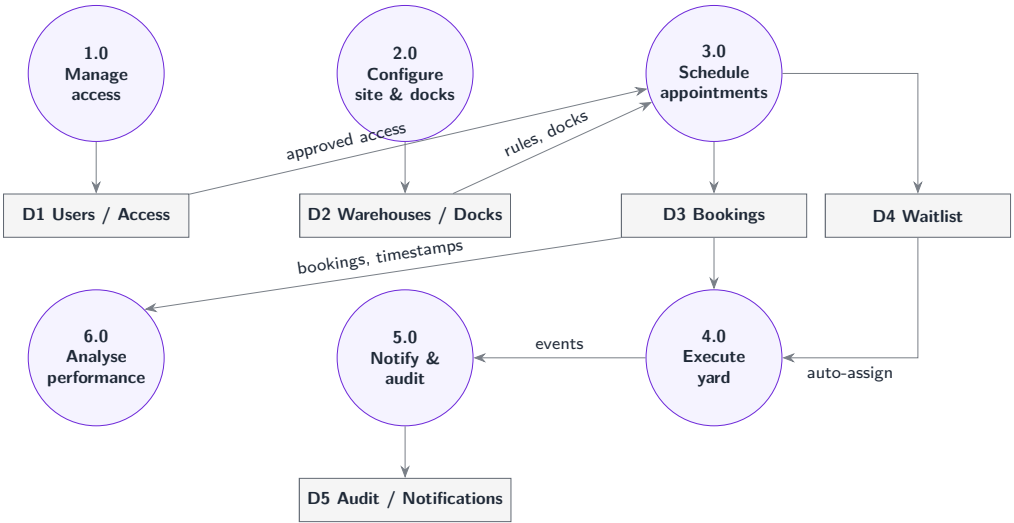


# Appointment Lifecycle (State Machine)

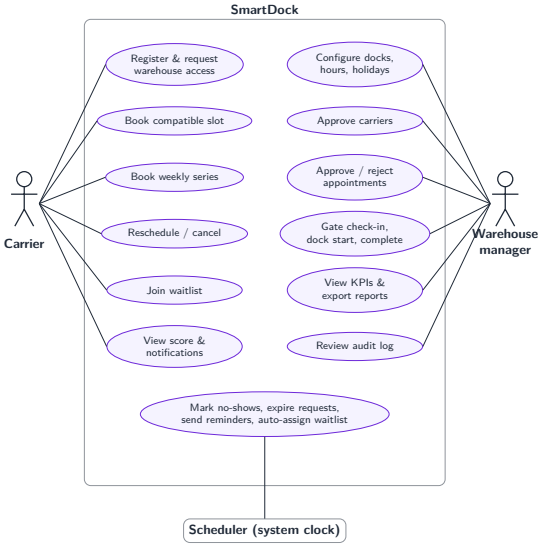



COMPLETED, CANCELLED, REJECTED and NO_SHOW are terminal: released capacity can never be silently re-occupied. Late / on-time is recorded at gate check-in; NO_SHOW is set automatically after the grace period.

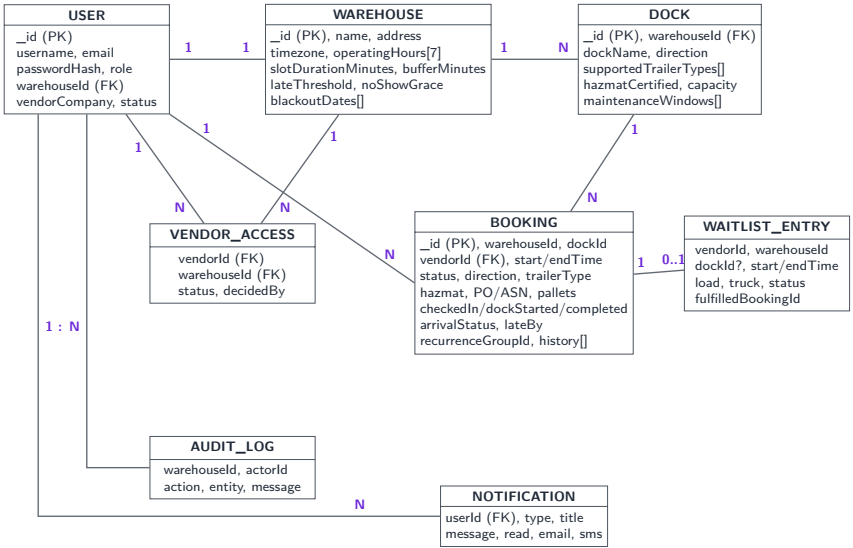
# Data Flow Diagram (Level 1)



# Use Case Diagram



# Entity–Relationship Diagram



# Technology Stack

| | |
|---|---|
| **Language** | TypeScript (client and server) |
| **Frontend** | React 19, Vite, Tailwind CSS, React Query, React Hook Form + Zod, jsPDF |
| **Backend** | Node.js 22, Express, Zod validation, JWT, bcrypt, rate limiting, Winston |
| **Database** | MongoDB Atlas (replica set), Mongoose — transactions, indexes, aggregations |
| **APIs** | REST/JSON (≈40 endpoints); Resend (e-mail), Twilio (SMS) |
| **Cloud & DevOps** | Render (Blueprint `render.yaml`, auto-deploy from GitHub), Atlas M0 |
| **Testing & tools** | In-memory replica set + 2 API instances (58 checks); Git, VS Code, Python, LaTeX |

# Modules

1. **Identity & access** — roles, per-warehouse approval
2. **Site & docks** — doors, equipment, hours, rules, holidays, maintenance
3. **Scheduling engine** — slot grid, availability, atomic reservation
4. **Booking** — wizard, weekly series, reschedule
5. **Waitlist** — FIFO auto-assignment
6. **Yard execution** — gate check-in, dock start, complete, no-show
7. **Carrier performance** — reliability score, tiers A–D
8. **Analytics** — KPIs, heatmap, CSV/PDF, gate sheet
9. **Notifications** — in-app, e-mail, SMS, reminders
10. **Audit trail** — who did what, when

# Implementation: Key Logic

**Slot grid** (site time zone, DST-safe)

$$s_k = o + k(L + B), \quad e_k = s_k + L \le c$$

$L$ = slot length, $B$ = buffer (30 + 15 ⇒ 08:00, 08:45, . . . )
**Compatibility**: direction matches (or both) ∧ trailer type supported ∧ (not hazmat ∨ certified)
**Arrival**: late if check-in − slot start > threshold (15 min)
**Background jobs** (every minute): no-shows, expiry, 24 h reminders, waitlist expiry

## Carrier reliability score

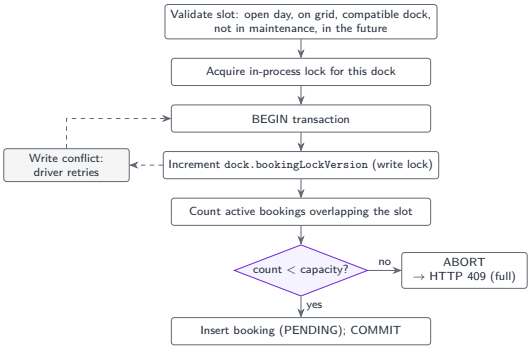
$$R = \frac{\text{on-time} + 0.5 \times \text{late}}{\text{arrivals} + \text{no-shows}} \times 100$$

A $\ge 90$ · B $\ge 75$ · C $\ge 50$ · D $< 50$

## Waitlist

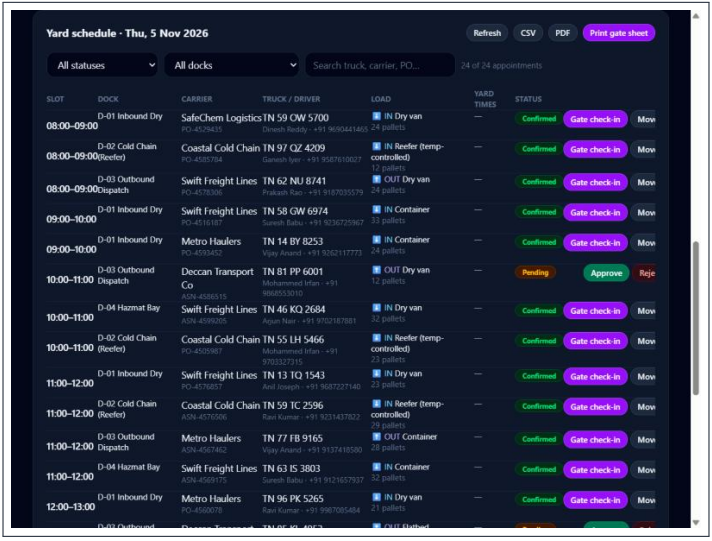
Capacity released (cancel / reject / reschedule) ⇒ oldest compatible waiting carrier booked automatically, inside the same transaction.

# Implementation: Concurrency-Safe Capacity Reservation

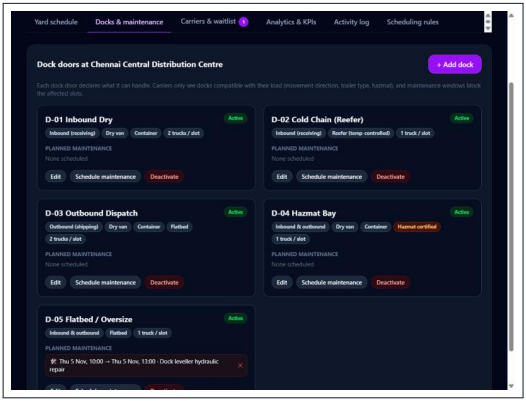


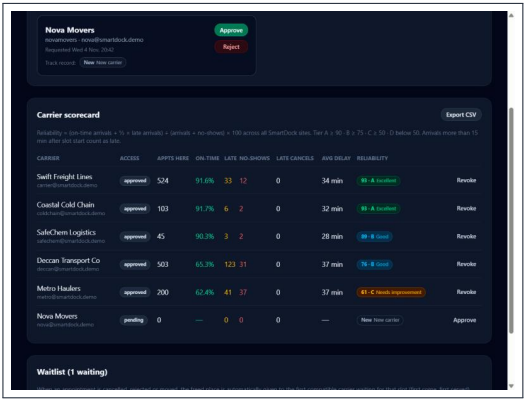

- Naive "count, then insert" lets two requests both see a free place (write skew)
- Even a transaction alone does not help: each reads its own snapshot
- SmartDock: every reservation writes the dock document first $\Rightarrow$ concurrent transactions conflict, the database retries them, the retry sees the booking
- Plus an in-process lock per dock
- Same routine for bookings, series, reschedules, waitlist

# Screens: Yard Schedule (Warehouse Manager)

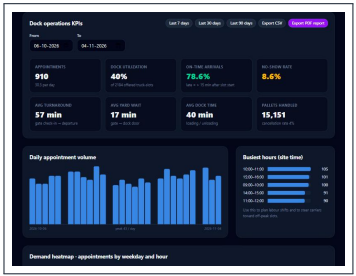


# Screens: Docks and Carrier Scorecard

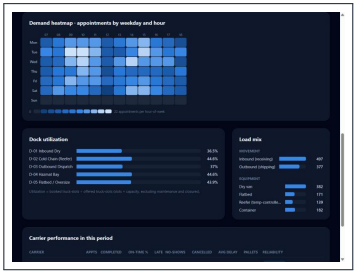


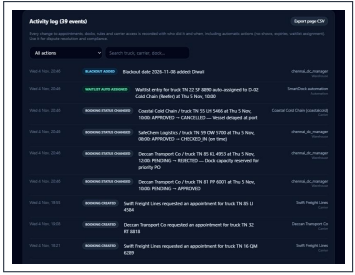
# Screens: Analytics, Rules and Audit




KPIs

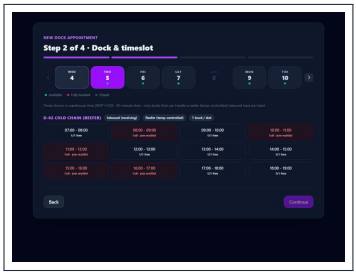



Heatmap & utilisation

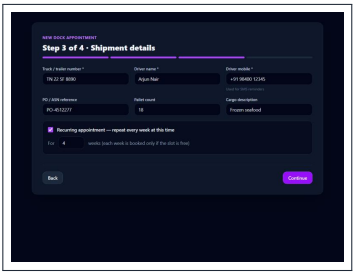



Activity log

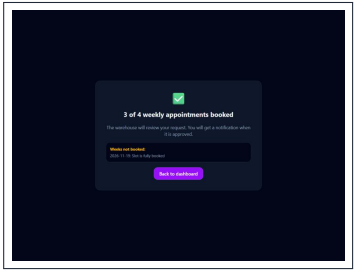
# Screens: Carrier Booking Wizard




Compatible docks only; waitlist on full slots




Truck, driver, PO/ASN, weekly series




3 of 4 weeks booked

# Live Demo

## Links

**Live:** `https://smartdock.onrender.com`
**Code:**
`https://github.com/RamaguruS/Dock`

Manager: `admin@smartdock.demo`
Carrier: `carrier@smartdock.demo`
Password: `Demo@123`

## Demo flow

1. Carrier books a reefer slot, weekly series; joins a waitlist
2. Manager approves → gate check-in → dock → complete
3. Docks, maintenance, rules, holidays
4. Analytics, PDF report, gate sheet, audit log
5. New carrier (Nova Movers) approval

# Testing and Validation

- 58 automated end-to-end checks on an in-memory **replica set** with **two** API servers — all pass
- Security & tenancy, capacity & concurrency, slot rules, dock matching, maintenance, holidays, lifecycle, series, waitlist, reschedule, reliability, audit, analytics
- Live smoke test on the production deployment

## Double-booking experiment

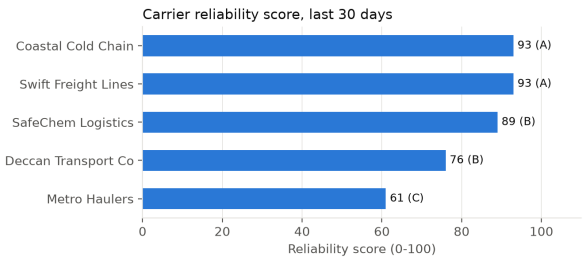
40 simultaneous requests, 2 servers

| | Cap. 1 | Cap. 3 |
|---|---|---|
| Naive | 15 | 13 |
| **SmartDock** | **1** | **3** |

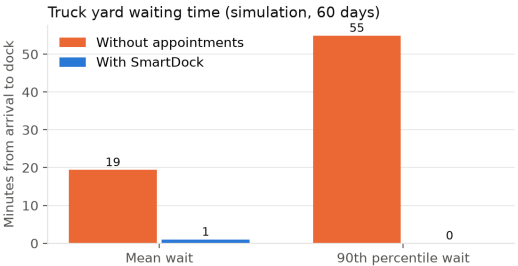
Live API (free tier): slot availability ≈1.3 s, yard schedule ≈1.6 s, 30-day analytics ≈4.3 s (server in USA, database in India)

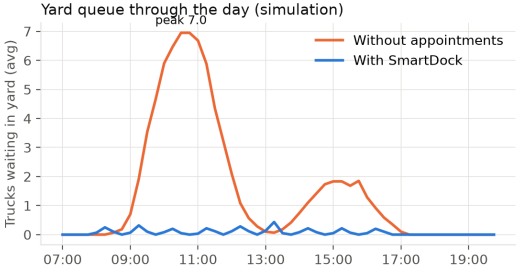
# Results: KPIs from the Deployed System (30 days, demo data)

| | |
|---|---|
| Appointments | 910 |
| Dock utilisation | 40.0% |
| On-time arrival rate | 78.6% |
| No-show rate | 8.6% |
| Avg. turnaround | 57 min |
| yard wait / dock time | 17 / 40 min |
| Pallets handled | 15,151 |
| Busiest hours | 10–11, 15–16 |



# Results: Before vs. After (Simulation, 6 docks, 60 trucks/day, 60 days)




| | Without appointments | With SmartDock |
|---|---|---|
| Mean yard wait | 19.4 min | **1.0 min** |
| Trucks waiting $>$ 30 min | 27.7% | **0.3%** |
| Peak yard queue | 7.0 trucks | **0.4 trucks** |

# Benefits and Real-World Applications

- **Cost**: less detention and overtime; labour rostered to known load
- **Visibility**: one live schedule for both sides
- **Speed**: self-service booking; waiting $\approx$19 $\rightarrow$ $\approx$1 min (simulated)
- **Traceability**: PO/ASN, timestamps, full audit trail
- **Reliability**: conflict-free, hazmat-safe; carrier accountability
- **Sustainability**: less idling near the site

## Applications

Retail & e-commerce DCs · cold-chain / pharma warehouses · chemical depots · 3PL multi-client sites · JIT manufacturing plants · container freight stations

# Conclusion

- Designed, built, tested and deployed a multi-tenant dock appointment and yard management system
- All objectives met: configurable scheduling, dock matching, guaranteed capacity, series, rescheduling, waitlist, yard events, reliability scores, KPIs, audit, notifications
- Verified by 58 automated checks and a concurrency experiment; live on the cloud
- A focused IT system at the warehouse–transport interface turns a phone-based process into a measured, collaborative one

# Future Enhancements

- **AI/ML**: predict no-shows and delays; smart overbooking; volume forecasting
- **Optimisation**: automatic door assignment; load-based slot length
- **IoT**: ANPR / RFID gate check-in, dock sensors, reefer temperature
- **Integration**: ERP/WMS/TMS (PO, ASN), webhooks, carrier API
- **Mobile**: driver app / PWA with directions and proof of delivery
- **Scale**: WebSockets, analytics caching, Asian region, super-admin console, CI

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# Thank you

Questions?

`https://smartdock.onrender.com` · `https://github.com/RamaguruS/Dock`

[Student Name] ([Register Number])